Understanding Spanning-Tree Protocol (STP) : A Practical Lab Guide

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1 Introduction

In a network of interconnected switches, data can get trapped in an endless loop, causing a **broadcast storm** that can bring the entire network to a halt. The **Spanning Tree Protocol (STP)** is the solution. It acts like a smart traffic controller, preventing these loops by logically *blocking* certain paths, ensuring that there is only one active, loop-free path between any two devices on the network.

This report provides a step-by-step breakdown of how STP works, from electing a network "root" to assigning port roles. We will use a five-switch topology (SW1 to SW5) to demonstrate these concepts.

2 The Network Topology

Our lab uses a physical topology of five interconnected switches. The objective is to analyze this network and determine which ports STP will block to create a loop-free topology.

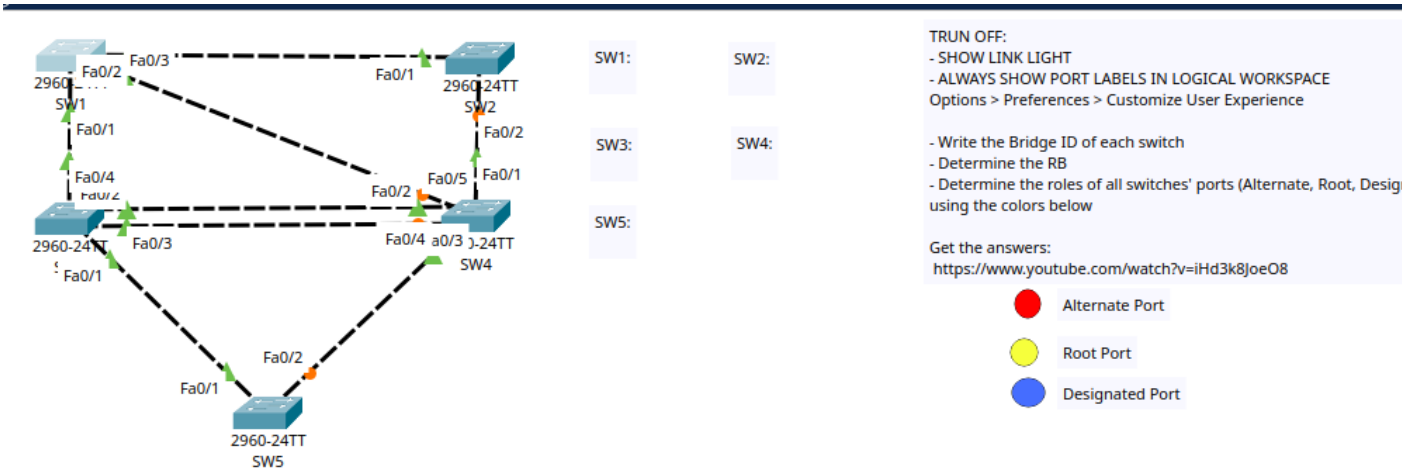


FIGURE 1 – Initial lab topology with five interconnected switches

To make the analysis visual, we will use a color-coded legend to mark the final role of each port :

- **Red** : Blocked Port (Alternate) — The port is disabled to prevent a loop.
- **Yellow** : Root Port — The port that provides the best path to the Root Bridge.
- **Blue** : Designated Port — The port responsible for forwarding traffic on a network segment.

3 How STP Works : The Four-Step Process

STP follows a strict election process to build a loop-free logical topology. The process consists of four key steps.

3.1 Step 1 : Elect the Root Bridge

The **Root Bridge** is the central point of the STP topology. It serves as the reference point for all path calculations.

- The Root Bridge is the switch with the **lowest Bridge ID (BID)**.
- If priorities are equal, the switch with the **lowest MAC address** becomes the Root Bridge.

In our topology :

- SW1, SW3, and SW4 all have the same default priority.
- Comparing MAC addresses : `SW3 < SW1 < SW4`.
- Therefore, **SW3 is elected as the Root Bridge**.

```
Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- -----
Fa0/1            Desg FWD 19        128.1    P2p
Fa0/2            Desg FWD 19        128.2    P2p
Fa0/3            Desg FWD 19        128.3    P2p
Fa0/4            Desg FWD 19        128.4    P2p

SW3#
```

FIGURE 2 – As SW3 is the Root Bridge, all its ports are in the *Designated* role and *Forwarding* state.

3.2 Step 2 : Select Root Ports

Every non-root switch must select exactly one **Root Port**. This is the port that provides the best path (lowest cost) back to the Root Bridge.

- **SW1 and SW5** : Directly connected to SW3, their direct links become their Root Ports.
- **SW4** : Has two links to SW3. Since both cost the same, we select the port with the **lowest Port ID** (interface number) on the neighboring switch. The port connected to SW3's `Fa0/2` is chosen.
- **SW2** : Has two possible paths to the Root Bridge : via SW1 or via SW4. Since both paths have the same cost, SW2 chooses the path through the switch with the **lower Bridge ID**. ($MACOFSW1 < MACOFSW4$), so SW2's port to SW1 becomes its Root Port.

```
SW3         Fas 0/2         178          S       2960        Fas 0/2
SW3         Fas 0/3         178          S       2960        Fas 0/3
SW4#
```

FIGURE 3 – Redundant links between SW3 and SW4 requiring tie-breaking

3.3 Step 3 : Select Designated Ports

On every network segment (link between two switches), exactly one **Designated Port** is selected. This port is responsible for forwarding traffic on that segment.

- The Designated Port is typically on the switch that has the **lowest cost** to reach the Root Bridge.
- If costs are equal, the switch with the **lower Bridge ID** wins.

In our topology :

— The ports on the opposite end of the root ports are always Designated Ports.

— **Segment SW4–SW1** : Both switches have equal cost. SW1 has a lower MAC address, so **SW1's port** becomes Designated.

— **Segment SW4–SW2** : SW4 has a better path to the Root Bridge, so **SW4's port** becomes Designated.

— **Segment SW4–SW5** : SW4 has the lower BID, so **SW4's port** becomes Designated.

3.4 Step 4 : Block Remaining Ports

Any port that is *neither* a Root Port *nor* a Designated Port is placed in a **Blocking** state. These blocked ports are called *Alternate Ports*. They remain inactive (blocking traffic) but are ready to become active if another link fails. This ensures a loop-free topology.

4 Final colored Topology

To confirm our manual analysis, we use Cisco's IOS command `show spanning-tree`. This command provides all the information we calculated theoretically.

```
Switch#show spanning-tree
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    4097
             Address     0030.A3C8.5498
             Cost        19
             Port        1(FastEthernet0/1)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    4097  (priority 4096 sys-id-ext 1)
             Address     0090.2192.9643
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------
Fa0/1            Root FWD 19        128.1    P2p
Fa0/2            Desg FWD 19        128.2    P2p
Fa0/3            Desg FWD 19        128.3    P2p
```

FIGURE 4 – Output of the `show spanning-tree` command confirming STP roles

After running the STP election process, the network converges to a stable, loop-free topology. The color-coded ports clearly show the final roles.

Figure 5 shows the final state :

— **Blue** : Designated Ports (mostly on SW3 and winning segments).

— **Yellow** : Root Ports (on SW1, SW2, SW4, and SW5 pointing toward the Root Bridge).

— **Red** : Blocked Ports (Alternate Ports on SW4, SW5, and SW2 that break the loops).

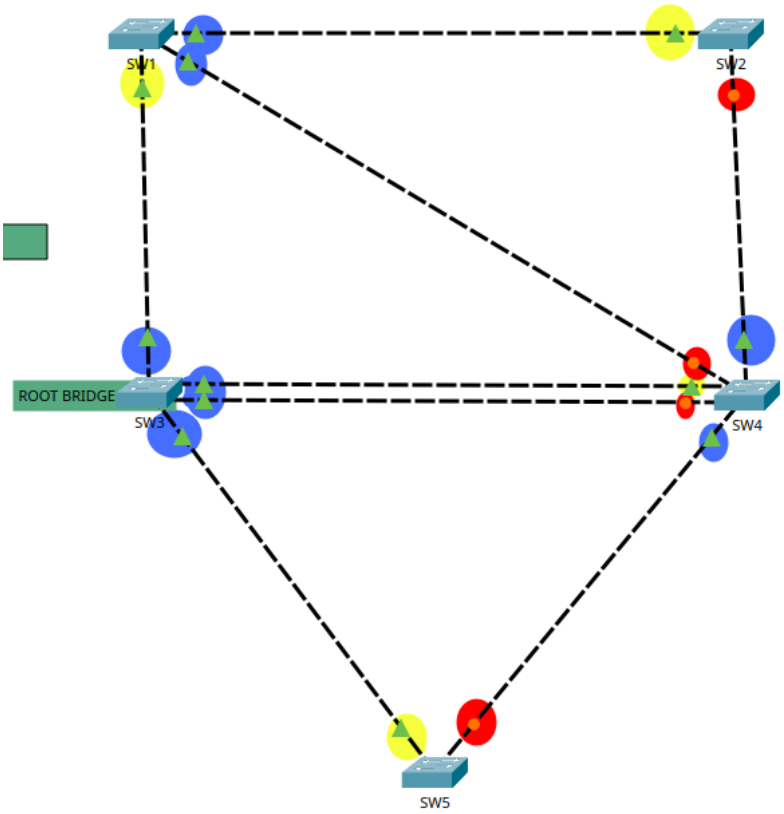


FIGURE 5 – Converged topology with all port roles assigned

5 Conclusion

This lab demonstrates the core mechanics of the Spanning Tree Protocol in a practical, easy-to-understand way. By following the four-step election process — electing a Root Bridge, selecting Root Ports, choosing Designated Ports, and blocking the remaining ports — STP successfully transforms a physically redundant (and potentially looping) network into a logically loop-free one.

Understanding STP is essential for any network engineer, as it forms the foundation of network redundancy and stability. This exercise confirms that our theoretical understanding matches the actual behavior of Cisco switches, validating the concepts behind one of the most important protocols in networking.